E-COMMERCE SALES DASHBOARD

Power BI Project Report

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1. Project Overview

1.1 Introduction

Online shopping creates a huge amount of sales data every single day. To make sense of this data, this project uses Power BI to build an interactive E-commerce Sales Dashboard. The dashboard turns raw, order-level data into simple and clear insights about revenue, profit, customer returns, and even environmental impact.

The dashboard gives a full picture of business performance by looking at sales, profit, orders, product returns, customer demographics (age, gender, location), payment methods, shipping methods, and sustainability metrics such as CO2 emissions and packaging waste. Instead of scrolling through thousands of rows of raw data, anyone using the dashboard can check the key numbers at a glance and use filters to explore further.

The dashboard was built with a simple, clean, dark-themed design so that even non-technical users can understand the numbers quickly and make faster decisions.

2. Project Objectives

The main goal of this project was to study e-commerce sales performance and present the findings through an interactive Power BI dashboard. The specific objectives include:

- Analyze overall sales, profit, and order volume.
- Track key performance indicators (KPIs) for quick, at-a-glance decisions.
- Identify the top-performing product categories.
- Understand customer return behaviour and the main reasons behind returns.
- Study customer buying patterns across age group, gender, and location.
- Compare payment method and shipping method preferences.
- Measure the environmental impact of orders using CO2 and packaging-waste data.
- Build an interactive dashboard with filters (slicers) for deeper, category-wise analysis.

3. Data Source

The dataset used in this project is a synthetic (simulated) e-commerce dataset named synthetic_ecomm.csv. It was created to reflect realistic online shopping behaviour, and it was imported into Microsoft Power BI using the Get Data feature. The data was checked for consistency before building any visuals.

3.1 Dataset Summary

Attribute	Description
Source	Synthetic e-commerce dataset (CSV file)
Format	CSV
Rows	5,000
Columns	22
Tool Used	Power BI

The dataset covers order details (Order ID, Product ID, User ID, Order Date), product details (Category, Price, Quantity), customer details (Age, Gender, Location), payment and shipping information, return details (Return Status, Return Reason, Days to Return), and sustainability figures (CO2 Emissions, CO2 Saved, Packaging Waste, Waste Avoided).

Order_ID	Product_ID	User_ID	Order_Date	Product_Category	Product_Price	Order_Quantity	Discount_Applied	Shipping_Method	Payment_Method	User_Age	User_Gen
ORD00186	PROD0249	USER0939	20 December 2024	Clothing	671.23	5	27.79	Standard	COD	54	Female
ORD00232	PROD0279	USER0872	02 October 2023	Clothing	1999.28	5	47.69	Standard	Wallet	29	Female
ORD00252	PROD0419	USER0974	10 December 2024	Clothing	370.12	5	29.32	Standard	Credit Card	65	Male
ORD00345	PROD0164	USER0208	04 November 2022	Clothing	526.53	5	26.25	Standard	Debit Card	41	Female
ORD00463	PROD0335	USER1661	01 May 2023	Clothing	178.2	5	21.69	Standard	COD	54	Female
ORD00480	PROD0390	USER1124	04 August 2023	Clothing	1124.4	5	21.03	Standard	Credit Card	41	Male
ORD00526	PROD0138	USER1772	26 October 2022	Clothing	1816.88	5	10.74	Standard	Wallet	31	Male
ORD00668	PROD0455	USER0432	03 March 2025	Clothing	718.67	5	21.5	Standard	Credit Card	30	Male
ORD00675	PROD0244	USER0060	09 January 2025	Clothing	1479.38	5	41.78	Standard	Debit Card	18	Female
ORD00703	PROD0220	USER1690	10 March 2023	Clothing	1786.18	5	14.62	Standard	Credit Card	32	Female
ORD00731	PROD0048	USER1951	16 January 2023	Clothing	1819.08	5	15.01	Standard	Credit Card	57	Female
ORD00752	PROD0290	USER1995	16 May 2024	Clothing	1732.64	5	10.28	Standard	COD	43	Male
ORD00942	PROD0083	USER1861	14 July 2024	Clothing	1789.05	5	7.55	Standard	COD	53	Female
ORD01060	PROD0362	USER1055	17 January 2025	Clothing	946.25	5	26.33	Standard	COD	49	Female
ORD01128	PROD0263	USER0212	11 January 2023	Clothing	981.88	5	44.69	Standard	COD	19	Female
ORD01211	PROD0493	USER0306	20 January 2023	Clothing	443.63	5	18.33	Standard	Debit Card	22	Female
ORD01251	PROD0251	USER1355	19 March 2023	Clothing	1835.7	5	20.55	Standard	Debit Card	29	Female
ORD01265	PROD0175	USER1898	20 October 2023	Clothing	1186.6	5	1.67	Standard	Debit Card	22	Male
ORD01332	PROD0302	USER0748	12 May 2023	Clothing	388.21	5	12.99	Standard	Credit Card	56	Female
ORD01339	PROD0334	USER0196	23 October 2024	Clothing	643.04	5	8.49	Standard	COD	61	Female
ORD01348	PROD0402	USER0482	02 November 2023	Clothing	1768.51	5	38.51	Standard	Wallet	34	Male
ORD01518	PROD0287	USER0627	01 February 2022	Clothing	1630.05	5	42.02	Standard	Debit Card	53	Male
ORD01572	PROD0207	USER1616	19 January 2023	Clothing	1239.67	5	12.22	Standard	Debit Card	22	Male
ORD01588	PROD0312	USER1516	28 July 2024	Clothing	1383.7	5	23.61	Standard	COD	21	Male
ORD01625	PROD0002	USER0830	05 December 2024	Clothing	217.91	5	41.23	Standard	COD	51	Male
ORD01812	PROD0348	USER1781	01 March 2022	Clothing	1272.11	5	5.39	Standard	Wallet	38	Female
ORD01919	PROD0270	USER0057	08 October 2023	Clothing	759.28	5	27.33	Standard	Debit Card	44	Female

A preview of the raw order-level data imported into Power BI.

4. Data Preparation & Modeling

Before building the dashboard, the dataset was prepared using Power Query Editor. The data preparation process involved:

- Importing the CSV dataset into Power BI.
- Checking column data types (dates, numbers, and text).
- Checking for missing or blank values.
- Formatting numerical fields such as price, order value, and profit.
- Creating calculated measures using DAX.
- Organizing all measures into a single '_measures' folder for easy access.

Since the dataset came from a single flat table, no table relationships were needed, which kept the data model simple and easy to work with.

4.1 DAX Measures Created

Measure	Purpose
Total Revenue	Sum of order value
Total Profit	Sum of profit / loss
Total Order	Count of orders
Avg Order Value	Average revenue per order
Return rate	Percentage of orders that were returned
returned Orders	Count of returned orders
Total Return Cost	Sum of the cost caused by returns
CO2 Emitted (kg)	Total CO2 released because of orders
CO2 Saved (kg)	Total CO2 saved through eco-friendly choices
CO2 Offset Ratio	CO2 saved divided by CO2 emitted
Waste Avoided (kg)	Total packaging waste avoided

5. Dashboard Development Approach

The dashboard was designed using a top-down approach so that users can understand business performance in seconds and then dig deeper if they want to.

The first section shows the most important KPIs — Total Revenue, Total Order, Average Order Value, Total Profit, Return Rate, and CO2 Saved — giving an instant snapshot of business health.

The second section looks at revenue and returns, showing which product categories earn the most and why customers send products back.

The third section studies customer behaviour — how revenue and orders change across gender, discount tier, age group, shipping method, payment method, month, and location.

The fourth section focuses on sustainability, using a CO2 Offset Ratio gauge to show how much of the CO2 produced by deliveries is being balanced out.

A Product Category slicer (Books, Clothing, Electronics, Home Appliances, Toys) sits at the top of the dashboard, letting users filter every visual by category with a single click.

6. Dashboard Analysis



Full view of the E-commerce Sales Dashboard built in Power BI.

6.1 KPI Cards

The KPI cards at the top give a quick, one-glance overview of business performance.

KPI	Purpose
Total Revenue	Total money earned from all orders (11.83M)
Total Order	Total number of orders placed (5K)
Avg Order Value	Average amount spent per order (2.37K)
Total Profit	Overall profit earned after costs (11.54M)
Return Rate	Share of orders that were returned (0.29, i.e. ~29%)
CO2 Saved (kg)	Total CO2 emissions saved through sustainable choices (5.33K)

6.2 Total Revenue by Product Category

A bar chart shows revenue earned by each product category. Clothing brings in the most revenue (about 4.2M), followed by Electronics (2.9M), Toys (1.9M), Books (1.8M), and Home Appliances (1.2M). This helps the business see which categories to promote or stock more of.

6.3 Returned Orders by Return Reason and Return Status

This bar chart looks only at returned orders and breaks them down by reason. 'Defective' items are the top reason for returns (382 orders), followed closely by 'Changed Mind' (379), 'Wrong Item' (348), and 'Size Issue' (341). This tells the business that product quality checks and accurate sizing/descriptions could reduce returns.

6.4 Total Revenue by User Gender and Discount Tier

A matrix table splits revenue by customer gender (Female / Male) and by discount tier (0-15%, 15-30%, 30-45%). Revenue is fairly close between genders, with Male customers in the 0-15% discount tier contributing slightly more (about 2.20M) compared to Female customers in the same tier (about 2.13M). This shows discounts are being used across both genders in a balanced way.

6.5 Total Order by Age Group

A bar chart shows how many orders come from each age group. The 55-64 and 35-44 age groups place the most orders (around 1.1K each), followed closely by 45-54 and 25-34 (around 1.0K each). Customers aged 65+ place far fewer orders, suggesting this group may need a different marketing approach.

6.6 Total Order by Shipping Method

This chart compares Next-Day, Standard, and Express shipping. Next-Day delivery is the most popular choice (about 2K orders), followed by Standard (1.7K) and Express (1.6K), showing that customers generally prefer faster delivery when it's available.

6.7 Total Order by Payment Method

A donut chart shows that orders are almost evenly split across the four payment methods — Credit Card (25.54%), COD (25.5%), Wallet (24.7%), and Debit Card (24.26%). No single payment method dominates, so the business should keep supporting all of them equally.

6.8 Total Order and Total Revenue by Month

A combination chart (bars for orders, a line for revenue) tracks performance across all twelve months. Both orders and revenue stay fairly steady through the year, with a slight dip toward the later months (October to December), which could point to a seasonal slowdown worth investigating.

6.9 Total Revenue by User Location

A bar chart compares revenue across different cities. Revenue is spread quite evenly across locations, with the top cities (such as City68, City35, and City10) each contributing around 0.17M. This suggests the customer base is well spread out rather than concentrated in one place.

6.10 CO2 Offset Ratio

A gauge visual shows a CO2 Offset Ratio of 0.42 (on a scale of 0 to 0.83). This means a solid portion of the CO2 produced by deliveries is currently being offset, but there is still room to improve packaging and delivery choices to push this ratio higher.

7. Key Skills & Techniques Demonstrated

Throughout this project, a mix of technical and analytical skills were applied.

Technical Skills	Analytical Skills
Power BI	Business Analysis
Power Query	KPI Development
DAX Measures	Trend Analysis
Data Cleaning	Customer Analysis
Data Visualization	Return & Loss Analysis
Dashboard Design	Sustainability Analysis
Interactive Slicers	Decision Support

8. Business Insights

The dashboard reveals several useful business observations:

- Clothing is the highest revenue-generating product category, followed by Electronics.
- 'Defective' products are the leading cause of returns, closely followed by customers simply changing their mind.
- The overall return rate is around 29%, which is fairly high and may be worth investigating further.
- Orders are spread almost equally across all four payment methods, so no single method needs to be prioritized over the others.
- Customers aged 35-64 are the most active buyers, while the 65+ age group orders the least.
- Next-Day delivery is the most preferred shipping method, showing customers value speed.
- Revenue is spread evenly across customer locations rather than concentrated in a few cities.
- The CO2 Offset Ratio of 0.42 shows decent progress on sustainability, with room to improve further.

9. Conclusion

This project shows how Power BI can turn raw, order-level e-commerce data into a clear and interactive business intelligence tool. By bringing together KPIs, charts, filters, and sustainability metrics into a single dashboard, it becomes much easier to monitor sales performance, understand customer behaviour, and track environmental impact.

The dashboard follows a clean, consistent, dark-themed design with interactive visuals that make it easy to use. It gives decision-makers clear insight into sales trends, product performance, return behaviour, and sustainability — supporting faster, data-driven business decisions.